Linear Regression

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Problem Formulation

What do we have?

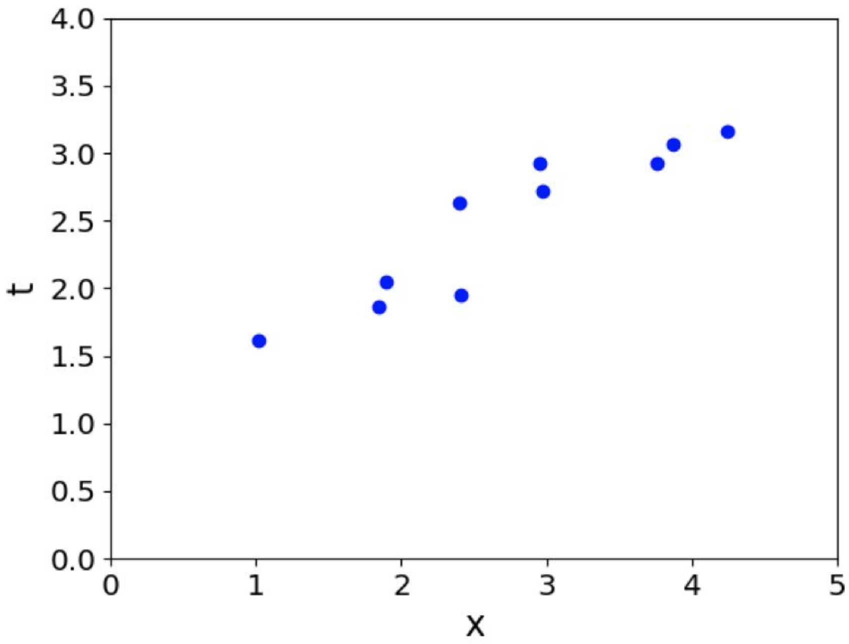
A training dataset comprising **N** observations $\{x_n\}$, where n = 1, . . . , N , together with corresponding labels $\{y_n\}$.

What is our goal?

To predict the value of **y** for a new value of **x**.

How do we do that?

We try to construct a function $\hat{y}(x)$ that predicts $\boldsymbol{y}$ for new inputs $\boldsymbol{x}$, or more generally, to model the predictive distribution p(y | x).



Simple Regression Model

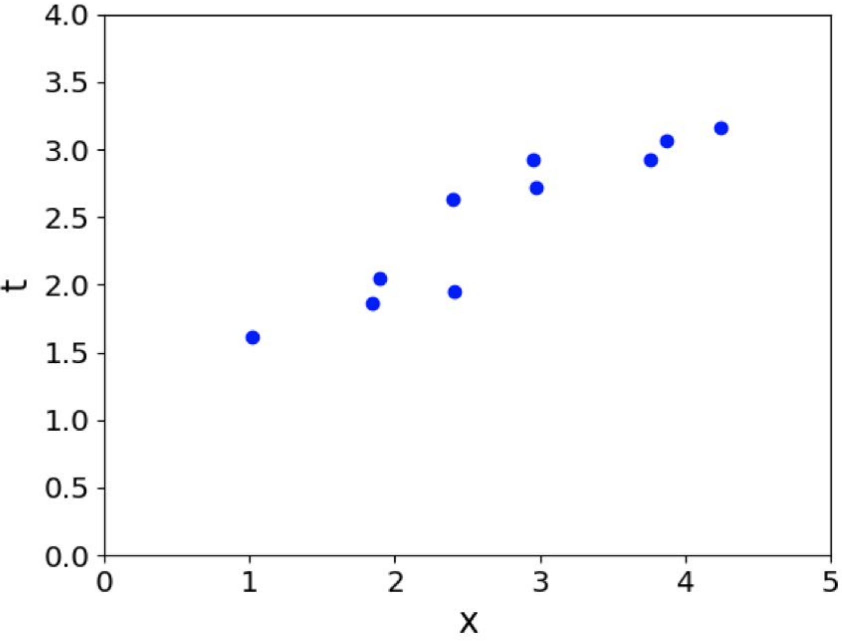
The simplest linear model for regression is:

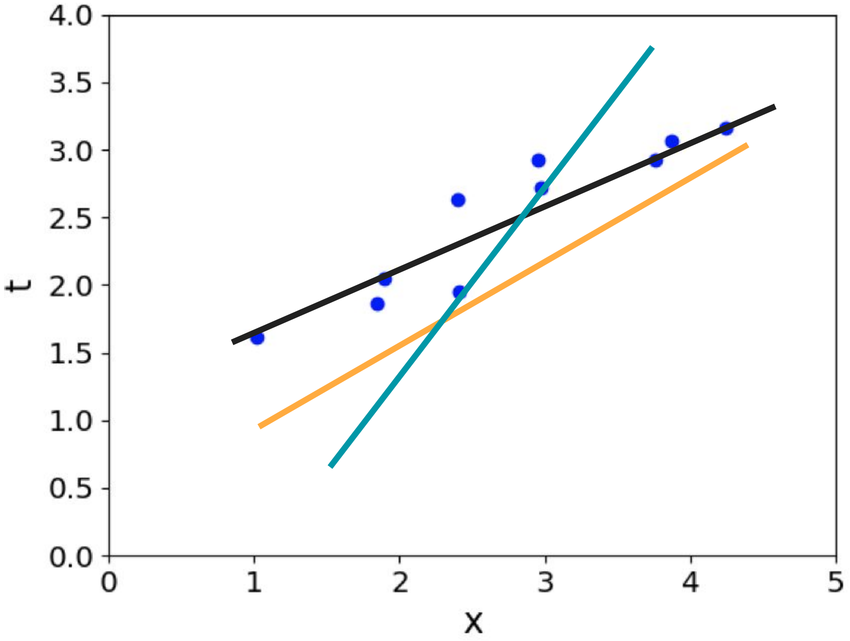
A linear combination of the input variables $\boldsymbol{x} = (x_1, \ldots, x_D)^T$:

$$y(\mathbf{x}, \mathbf{w}) = w_0 + w_1 x_1 + \ldots + w_D x_D$$

The key property of this model is that **it is a linear function of the parameters $w_0, \ldots, w_D$**

Illustration





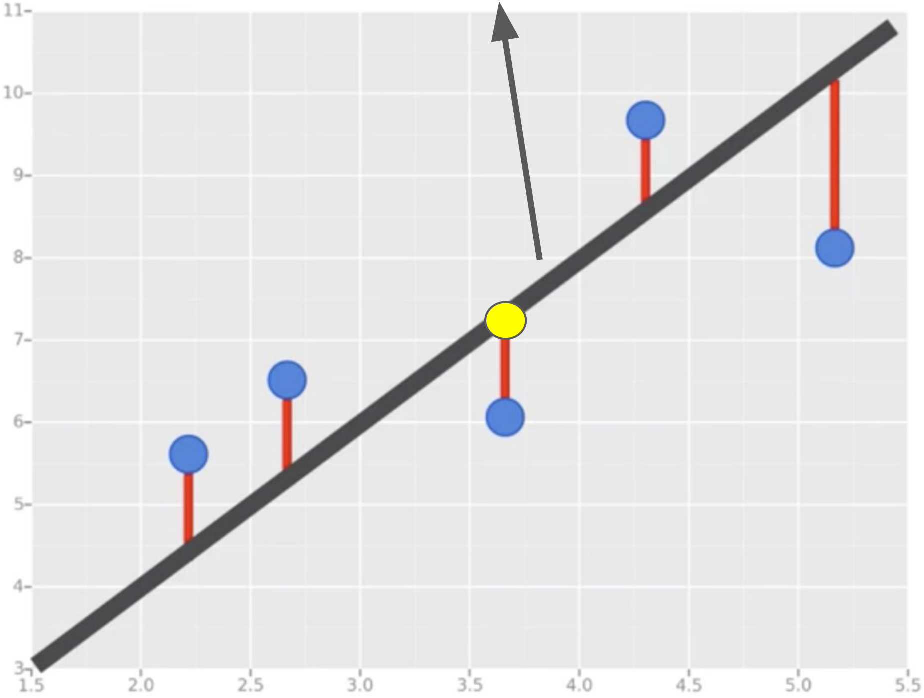
Least squares method

- Minimizing the sum of squares of the residuals
- **Residual** - difference between the observation (actual y-value) and the fitted line i.e. $y_i - \hat{y}$
- **Slope** - (change in y) / (change in x)
- **Intercept** - value of y when x=0

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

intercept slope

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$



MSE

Mean Squared Error (MSE) :

Calculates the squared difference between actual and predicted values. By squaring the difference, we avoid the cancellation of negative terms and it is the benefit of MSE.

$$MSE = \underbrace{\frac{1}{N}}_{\text{average over all results}} \sum_{i=1}^{N} (\underbrace{y_i}_{\text{true } y} - \underbrace{\hat{y}_i}_{\text{estimate of } y})^{\overset{\text{makes result quadratic}}{2}}$$

Advantages:

- The graph of MSE is differentiable, so you can easily use it as a loss function.

Disadvantages:

- The value you get after calculating MSE is a squared unit of output.
- If you have outliers in the dataset then it penalizes the outliers most and the calculated MSE is bigger. It is not Robust to outliers.

1D Linear regression

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

1) Take partial derivatives w.r.t. $\hat{\beta}_0$ and $\hat{\beta}_1$
2) Set the partial derivatives equal to 0
3) Solve the resulting equations for $\hat{\beta}_0$ and $\hat{\beta}_1$

1D Linear regression

$$\sum_{i=1}^{N}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$\frac{\partial}{\partial\hat{\beta}_0}\sum_{i=1}^{N}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^{N}\frac{\partial}{\partial\hat{\beta}_0}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^{N} 2 * (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) * (-1) =$$

$$= -2 * \sum_{i=1}^{N}(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))$$

$$\frac{\partial}{\partial\hat{\beta}_1}\sum_{i=1}^{N}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^{N}\frac{\partial}{\partial\hat{\beta}_1}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^{N} 2 * (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) * (-x_i) =$$

$$= -2 * \sum_{i=1}^{N} x_i * (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))$$

1D Linear regression

$$\frac{\partial}{\partial \hat{\beta}_0} \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = -2 * \sum_{i=1}^{N} (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)) = 0$$

$$\sum_{i=1}^{N} (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)) = 0$$

$$\sum_{i=1}^{N} y_i - \sum_{i=1}^{N} \hat{\beta}_0 - \sum_{i=1}^{N} \hat{\beta}_1 x_i = 0$$

$$\sum_{i=1}^{N} y_i - N\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^{N} x_i = 0$$

$$N\hat{\beta}_0 = \sum_{i=1}^{N} y_i - \hat{\beta}_1 \sum_{i=1}^{N} x_i$$

$$\boxed{\hat{\beta}_0 = \frac{\sum_{i=1}^{N} y_i - \hat{\beta}_1 \sum_{i=1}^{N} x_i}{N}}$$

1D Linear regression

$$\frac{\partial}{\partial \hat{\beta}_1} \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = -2 * \sum_{i=1}^{N} x_i * (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)) = 0$$

$$\sum_{i=1}^{N} x_i * (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)) = 0$$

$$\sum_{i=1}^{N} (x_i y_i - \hat{\beta}_0 x_i - \hat{\beta}_1 x_i^2) = 0$$

$$\sum_{i=1}^{N} x_i y_i - \hat{\beta}_0 \sum_{i=1}^{N} x_i - \hat{\beta}_1 \sum_{i=1}^{N} x_i^2) = 0 \longleftarrow \hat{\beta}_0 = \frac{\sum_{i=1}^{N} y_i - \hat{\beta}_1 \sum_{i=1}^{N} x_i}{N}$$

1D Linear regression

$$\sum_{i=1}^{N} x_i y_i - \frac{(\sum_{i=1}^{N} y_i - \hat{\beta_1} \sum_{i=1}^{N} x_i) * \sum_{i=1}^{N} x_i}{N} - \hat{\beta_1} \sum_{i=1}^{N} x_i^2) = 0$$

$$\sum_{i=1}^{N} x_i y_i - \frac{1}{N} \sum_{i=1}^{N} y_i \sum_{i=1}^{N} x_i + \frac{\hat{\beta_1}}{N} (\sum_{i=1}^{N} x_i)^2 - \hat{\beta_1} \sum_{i=1}^{N} x_i^2 = 0$$

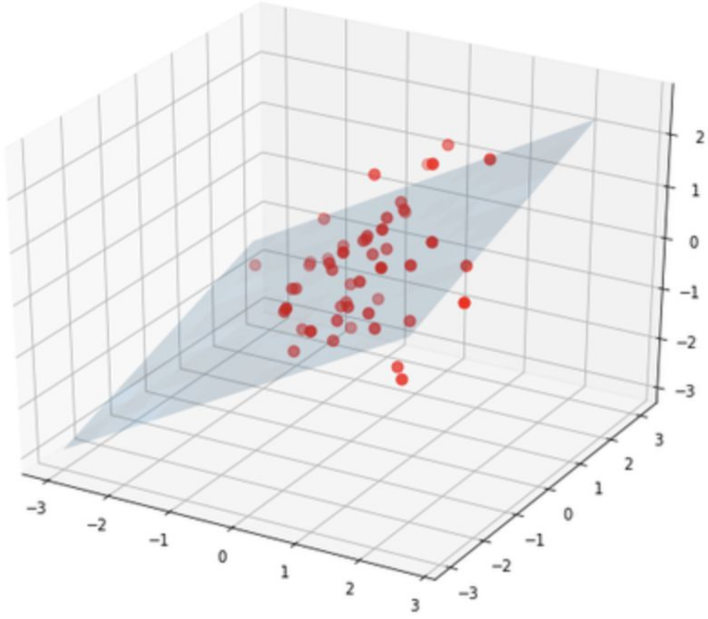
$$\sum_{i=1}^{N} x_i y_i - \frac{1}{N} \sum_{i=1}^{N} y_i \sum_{i=1}^{N} x_i = \hat{\beta_1} \sum_{i=1}^{N} x_i^2 - \frac{\hat{\beta_1}}{N} (\sum_{i=1}^{N} x_i)^2$$

$$\sum_{i=1}^{N} x_i y_i - \frac{1}{N} \sum_{i=1}^{N} y_i \sum_{i=1}^{N} x_i = \hat{\beta_1} (\sum_{i=1}^{N} x_i^2 - \frac{1}{N} (\sum_{i=1}^{N} x_i)^2)$$

$$\boxed{\hat{\beta_1} = \frac{\sum_{i=1}^{N} x_i y_i - \frac{1}{N} \sum_{i=1}^{N} y_i \sum_{i=1}^{N} x_i}{\sum_{i=1}^{N} x_i^2 - \frac{1}{N} (\sum_{i=1}^{N} x_i)^2}}$$

Multivariate linear regression

Suppose we have **n** data points of **k** dimensions - each data point is described with **k** features. A general multivariate model can be written as:



Multivariate linear regression

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_k x_{ik} + u_i \qquad \text{for } i = 1, \ldots, n.$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \ldots & x_{1k} \\ 1 & x_{21} & x_{22} & \ldots & x_{2k} \\ \\ 1 & x_{n1} & x_{n2} & \ldots & x_{nk} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \\ \beta_k \end{bmatrix} + \begin{bmatrix} u_1 \\ u_2 \\ \\ u_n \end{bmatrix}$$

n x 1   n x (k+1)   (k+1) x 1   n x 1

$$Y = X\beta + u \qquad \boldsymbol{u} = \boldsymbol{Y} - \boldsymbol{X}\beta$$

Error term: how far is the actual y from regression line

Multivariate linear regression

$$\min_{\beta} \quad u'u = (Y - X\beta)'(Y - X\beta)$$

$$u'u = (Y' - \beta'X')(Y - X\beta)$$

$$= Y'Y - \beta'X'Y - Y'X\beta + \beta'X'X\beta$$

$$= Y'Y - 2\beta'X'Y + \beta'X'X\beta$$

$$\underbrace{\underbrace{\underset{1\times n}{Y'}\;\underset{n\times(k+1)}{X}}_{1\times(k+1)}\;\underset{(k+1)\times 1}{\beta}}_{1\times 1}$$

Transpose of a scalar is equal to itself

$$Y'X\beta = (Y'X\beta)' = \beta'X'Y$$

Multivariate linear regression

$$u'u = Y'Y - 2\beta'X'Y + \beta'X'X\beta$$

$$\frac{\partial(u'u)}{\partial\beta} = -2X'Y + 2X'X\beta$$

$$-2X'Y + 2X'X\beta = 0$$

$$X'X\beta = X'Y$$

$$(X'X)^{-1}(X'X)\beta = (X'X)^{-1}X'Y$$

$$\boxed{\beta = (X'X)^{-1}X'Y}$$

Multivariate Linear Regression

Why is this a minimum?

Compute Hessian:

$$\nabla^2 J = 2/N\ X^T X$$

For any vector z:

$$z^T (X^T X) z = (Xz)^T (Xz) = ||Xz||^2 \geq 0$$

Therefore:

$X^T X$ is positive semidefinite

If columns of X are linearly independent:

$X^T X$ is positive definite

Multivariate Linear Regression

Geometric interpretation

We minimize:

$$\min_w ||y - Xw||^2$$

Define:

$$y\hat{} = Xw$$

Notice:

$$y\hat{} \in Col(X)$$

(the column space of X)

Multivariate Linear Regression

Geometric interpretation

Key Observation

We are searching for:

$$\hat{y} \in \mathrm{Col}(X)$$

that is closest to y.

Therefore:

Linear regression = **Orthogonal projection of y onto $\mathrm{Col}(X)$.**

Multivariate Linear Regression

Geometric interpretation

Define residual: $r = y - Xw$

At the minimum: $X^T r = 0$

Substitute:

$$X^T X w = X^T y$$

This is the Normal Equation.

Geometric Meaning: The residual vector is orthogonal to every column of X.

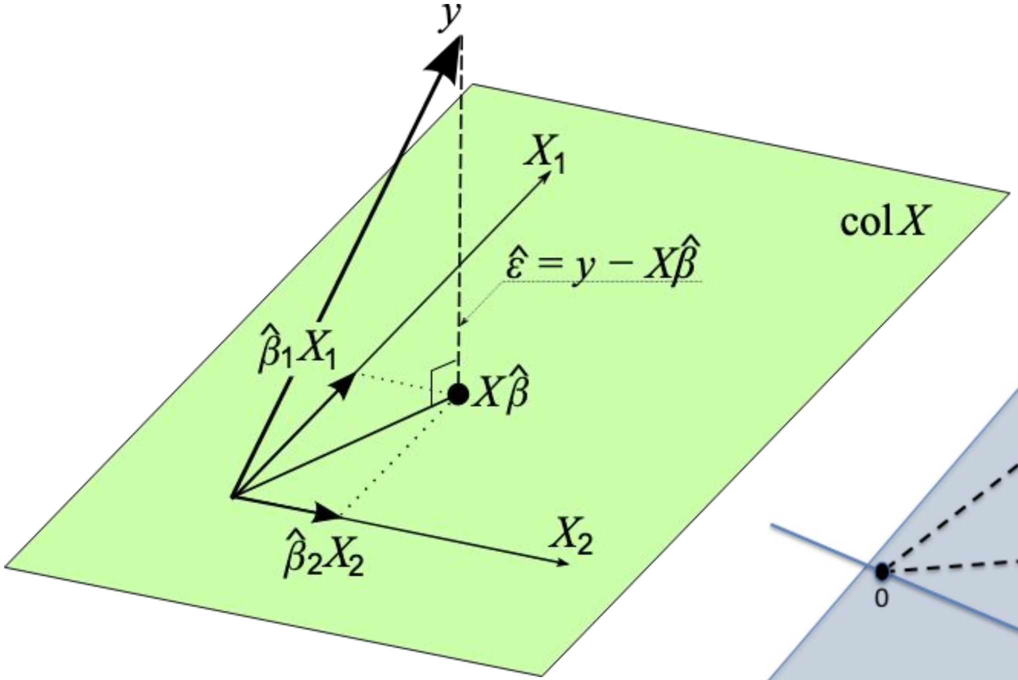
$$r \perp \mathrm{Col}(X)$$

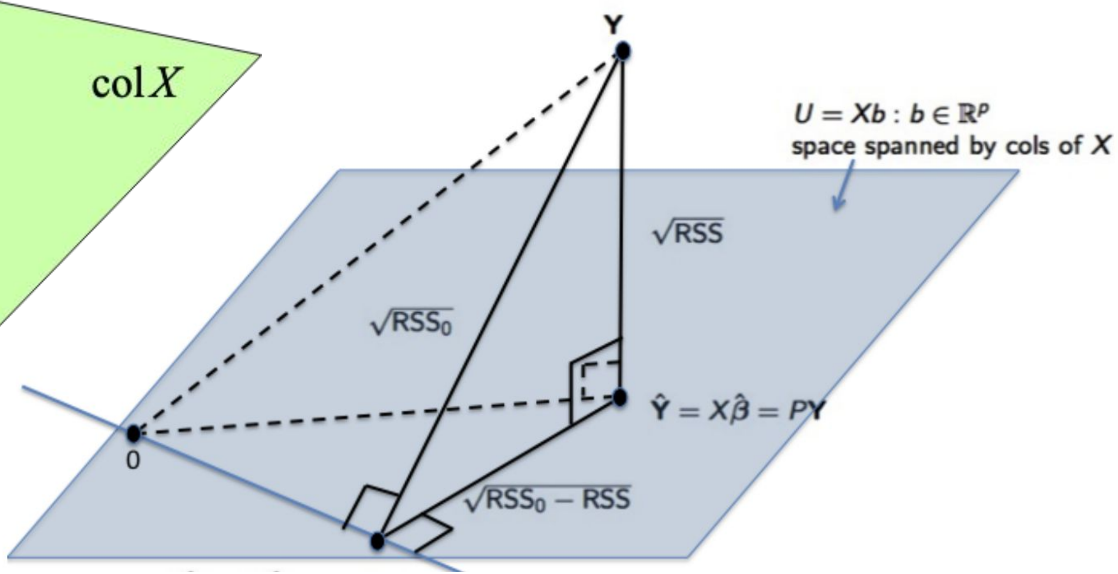
Interpretation: The error vector is perpendicular to the subspace.

That is exactly the condition for orthogonal projection.

Multivariate Linear Regression

Geometric interpretation





Multivariate Linear Regression

When X^TX Is Not Invertible

X^TX is singular if X is **rank deficient.**

This occurs when:

- Some features are linear combinations of others
- Two columns are identical
- $D>N$
- Perfect multicollinearity

$$\mathrm{rank}(X)<D$$

Multivariate Linear Regression

When X^TX Is Not Invertible

Normal equation:

$$(X^TX)w = X^Ty$$

If X^TX is singular:

$$(X^TX)^{-1} \text{ does not exist}$$

So we cannot compute:

$$w = (X^TX)^{-1}X^Ty$$

Multivariate Linear Regression

Does the minimum disappear?

No.

The optimization problem is still:

$$\min_w \|y - Xw\|^2$$

The loss is still convex.

A global minimum still exists.

But it is **not unique**.

Multivariate Linear Regression

Does the minimum disappear?

If $z \in \mathrm{Null}(X)$, then:

$$X(w+z)=Xw$$

So:

$$||y-X(w+z)||^2=||y-Xw||^2$$

There are infinitely many minimizers.

The Moore–Penrose Pseudoinverse

Instead of using:

$$w=(X^{T}X)^{-1}X^{T}y$$

we define:

$$w^{*}=X^{+}y$$

where X^{+} is the Moore−Penrose pseudoinverse.

The Moore–Penrose Pseudoinverse

Definition

Let $X \in \mathbb{R}^{m \times n}$.

The Moore–Penrose pseudoinverse of X, denoted X^+, is the unique matrix $X^+ \in \mathbb{R}^{n \times m}$ satisfying:

$$
\begin{aligned}
&(1) \quad XX^+X = X \\
&(2) \quad X^+XX^+ = X^+ \\
&(3) \quad (XX^+)^T = XX^+ \\
&(4) \quad (X^+X)^T = X^+X
\end{aligned}
$$

Interpretation of the Four Conditions

- (1) Consistency on the column space
- (2) Consistency on the row space
- (3) XX^+ is symmetric (projection onto $\mathrm{Col}(X)$)
- (4) X^+X is symmetric (projection onto $\mathrm{Row}(X)$)

Multivariate Linear Regression

What does it do?

It solves exactly:

$$\min_w ||y - Xw||^2$$

It does **not** approximate.

It finds an exact least-squares minimizer.

Multivariate Linear Regression

Among infinitely many solutions...

The pseudoinverse chooses:

$$w^{*}=\mathrm{argmin}_{w}||w|| \text{ subject to minimizing } ||y-Xw||^{2}$$

It gives the **minimum-norm solution**.

Geometric Interpretation

- The loss is flat along the null space.
- All solutions differ by vectors in $\mathrm{Null}(X)$.
- The pseudoinverse picks the solution orthogonal to the null space.

Role of the bias term

Insight into the role of the bias:

$$w_0 = \bar{y} - \sum_{j=1}^{D} w_j \bar{x}_j$$

The bias w_0 compensates for the difference between the averages (over the training set) of the target values and the weighted sum of the averages of the input values.

Residual Sum of Squares

The sum of the squares of residuals are equal to

$$\mathrm{RSS} = \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

$$\mathrm{MSE} = \frac{1}{N}\mathrm{RSS}$$

- Smaller RSS → better fit.
- But more features → smaller RSS always.

Model evaluation

- **R-squared** - proportion of variance explained (0,1)
- What is a good R-squared value? - hard to say
- More features - higher R-square => not a reliable approach for choosing the best model

$$Unexplained\ variance = \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$$

$$Total\ variance = \sum_{i=1}^{n} (y_i - avg(y))^2$$

$$R^2 = 1 - \frac{Unexplained\ variance}{Total\ variance}$$